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ORIGINAL PAPER

Increasing the robustness of phenological models for *Vitis vinifera* cv. Chardonnay

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Abstract Phenological models are important tools for planning viticultural practices in the short term and for projecting the impact of climate change on grapevine (*Vitis vinifera*) in the long term. However, the difficulties in obtaining phenological models which provide accurate predictions on a regional scale prevent them from being exploited to their full potential. The aim of this work was to obtain a robust phenological model for *V. vinifera* cv. Chardonnay. During calibration of the sub-models for budburst, flowering and veraison we implemented a series of measures to prevent overfitting and to give greater physiological meaning to the models. Among these were the use of experimental information on the response of Chardonnay to forcing temperatures, restriction of parameter space into physiologically meaningful limits prior to calibration, and simplification of the previously selected sub-models. The resulting process-based model had good internal validity and a good level of accuracy in predicting phenological events from external datasets. Model performance was especially high for the prediction of flowering and veraison, and comparison with other models confirmed it as a better predictor of phenology, even in extremely warm years. The modelling study highlighted a different phenological behaviour at the only mountain station, Cembra. We hypothesised that phenotypical plasticity could lead to growth rates adapting to a lower mean temperature, a mechanism not usually accounted for by phenological models.

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Introduction

Phenological models have a wide range of applications in viticulture. Among these are a more efficient planning of harvest and viticultural practices (Williams et al. 1985), the forecasting of plant vulnerability to pest attacks and the selection of cultivars to be grown in new areas (Gladstones 1992; Orlandini and Mancini 1999; Bindi et al. 1997; Oliveira 1998). In addition, phenological models applied to climate change scenarios can be used to project the impact of climate changes on grapevine (*Vitis vinifera*) (Moriondo and Bindi 2007). However, in order to obtain reliable extrapolations of global change scenarios, models have to be spatially robust (Chuine 2000; García de Cortázar-Atauri et al. 2009) and able to maintain good performance in extreme years.

The models of phenology for grapevine currently in use are based on degree days or Winkler degrees (Amerine and Winkler 1944), calculated by summing temperatures above a threshold (10°C for Winkler degrees). Unfortunately, these models usually have only local validity and are applicable to a narrow range of climatic conditions, similar to those used for calibration (García de Cortázar-Atauri et al. 2009). This is probably due to the fact that the models do not describe adequately the physiology underlying the phenological development of grapevine. Factors not considered by the models may, in fact, change from site to site, so that site-specific estimation of model parameters is required.

For example, simple degree-day models do not take into account the effect of chilling on dormancy release for the

prediction of budburst. Yet lack of chilling temperatures could result in incomplete or irregular budburst (Lavee and May 1997) or the necessity for chemical treatments, such as hydrogen cyanamide (Botelho et al. 2007). Hence, the inclusion of dormancy is important if the model is to be applied to climate contexts warmer than the present, as is the case with climate change scenarios. In addition, models for late summer phenological events, such as veraison or ripening, often fail to predict prior events such as flowering or budburst, so the opportunity to re-tune the simulation as the growing season progresses is lost.

To prevent these shortcomings, a more process-based representation of the model system is necessary. Although some aspects of the physiology of phenological development remain unclear (Lavee and May 1997), it is possible to develop models which are more biologically realistic than the current ones. This can be achieved by integrating existing knowledge about the responses of phenological development to environmental drivers with hypotheses on missing relationships. The resulting framework is more complex than a purely empirical model but contains more information on the processes which lead to a defined phenophase and allows a deeper insight into the system, besides providing grounds for the formulation of new hypotheses (Thornley and Johnson 1990). A consequence of this additional complexity is an increase in the number of parameters defining the model. If these parameters are to be estimated by adjusting the model to a dataset, measures to prevent model overfitting are of the foremost importance (Janssen and Heuberger 1995).

The aim of this study is to obtain a regionally robust model which predicts budburst, flowering and veraison in *V. vinifera* cv. Chardonnay. This aim is pursued through calibration of one sub-model for each stage. In an attempt to give the model good regional validity, we have given more importance to model stability, even if this meant that prediction performance was not as high as in site-specifically calibrated models developed for local use.

Materials and methods

Phenological cycle of grapevine

The cycle of grapevine development can be divided into four major phases:

(1) Endodormancy, broken by exposure to a period of chilling temperatures (Pouget 1963) and followed by ecodormancy, broken by exposure to warm, forcing temperatures; this stage ends with budburst.
(2) From budburst to flowering, the vegetative organs and floral parts develop. The reserves built up during the previous growing season are mobilised by the plant for the start of spring growth.
(3) From flowering to veraison, the plant actively accumulates biomass. Berries are actively growing through cell division and enlargement, but stay green and hard.
(4) From veraison to harvest, vegetative growth slows down, berries accumulate large amounts of sugars, while their acid content decreases, and they start to soften and undergo colour change. The ripening process of the grapes takes place.

Phenological and meteorological data

The phenological series used referred to the phases of budburst (BBCH 07), flowering (BBCH 65) and veraison (BBCH 85) of Chardonnay. We considered five series in total, three of which were from the southern alpine province of Trentino (San Michele, Mezzocorona and Cembra), one from Piedmont (Carpeneto) and one from Veneto (Conegliano) (Fig. 1). The Conegliano dataset consisted of two series from two villages close to each other and at similar altitudes, Susegana (data from 1969 to 2000) and Spresiano (data from 2001 to 2007). The datasets and stations are described in Table 1.

Daily mean temperatures were obtained from weather stations near the sites. In order to work on complete temperature series, missing data were reconstructed with regression techniques on values recorded at the closest stations.

Preliminary study and model selection

Exploratory regression analyses were performed to select the most important climatic factors affecting phenological development and to create a set of candidate phenological models to be fitted to the datasets. Multiple regression was thus carried out for the sites where meteorological variables other than temperature were available. Daily solar radiation and daily precipitation were available for San Michele and daily precipitation for Cembra and Carpeneto. The multiple regression models were tested for both main and interaction effects between meteorological agents aggregated as monthly means preceding the date of the phenophase. The model tested for San Michele was: day of phenophase versus temperature x precipitation x solar radiation, while the model tested for Cembra and Carpeneto was: day of phenophase versus temperature x precipitation. These models were simplified to their significant terms by stepwise selection. However, negligible effects could be attributed to solar radiation and precipitation, so we did not include any relationships based on these factors in the phenological models tested in the preliminary study. Thus,

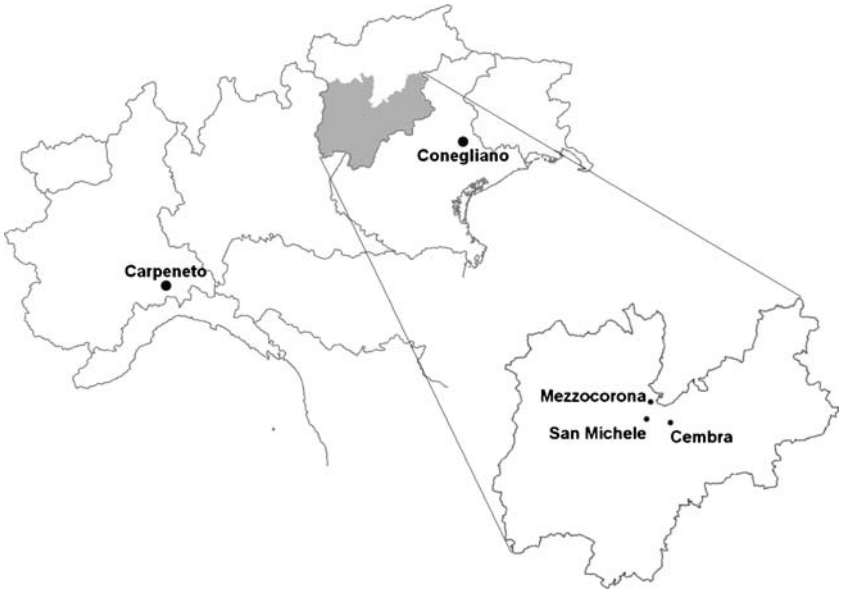


Fig. 1 Location of the sites of provenance of the phenological data series used in the study in northern Italy including Trentino province (*enlargement*)

the set of candidate models fitted to the available datasets were based solely on temperature and photoperiod (in terms of starting date for forcing unit accumulation). The models considered, seven in total, differed in their responses to temperature (linear or sigmoidal) and in the periods of action of temperature on phenological development. For budburst, we applied a general model which took into account the action of chilling temperatures ("Unified Model" for budburst in trees, proposed by Chuine 2000) as well as models which took into account only the accumulation of heat units, either through a sigmoidal or a linear function. For flowering and veraison, we considered only heat accumulation, which was calculated either from a calibrated calendar date or from the previous phenophase. The candidate models, which required the input of mean daily temperatures, are described in more detail in Table 2.

The goodness of fit (internal validity) of each model was measured by the portion of explained variation in the data and the mean absolute error (MAE). The portion of variation explained by the model (R_m^2) is defined as:

$$R_m^2 = \frac{SS_{tot} - SS_{res}}{SS_{tot}}$$

where SS_{tot} is the sum of the squared deviations of the observations from their mean, and SS_{res} is the sum of squared residuals of the model fit. This measure can yield negative values if the sum of squared model residuals is higher than the sum of the squared deviations.

When two models with a different number of parameters were compared, *F* tests were performed to establish which model was significantly better, taking into account the SS_{res} and the number of parameters (Motulsky and Christopoulos 2004). Where the two models compared did not differ significantly, the model with the lowest number of parameters was preferred.

After this preliminary model fit and evaluation, the models that accounted for more variation in the data were

Table 1 Elevation of the monitored sites, period of phenological monitoring, mean, standard deviation (SD) and number of observations (*n*) available for each phenophase. NA: no data available

Station	Elevation (m)	Site morphology	Monitored period	Budburst			Flowering			Veraison		
				n	Mean	SD	*n*	Mean	SD	*n*	Mean	SD
Conegliano	100	Foothills	1969–2007	39	103	7.4	39	155	8.4	39	218	9.1
San Michele all'Adige	228	Valley bottom	1987–2000	14	100	8	14	158	5	14	221	6.6
Mezzocorona	219	Valley bottom	1986–2006	21	103	7.9	20	153	5	20	215	6.9
Cembra	550	Mountain slopes	1988–2007	NA			20	160	8.1	12	225	9.2
Carpeneto	329	Hills	1987–2007	11	103	8.1	11	154	6.2	11	213	5.7

Table 2 Phenological models fitted to the phenological series from San Michele, Mezzocorona, Cembra and Carpeneto, and predicted phenophases

Model	Name and description	No. of parameters	Equation	Budburst	Flowering	Veraison
1	Unified Model for Budburst (Chuine 2000). Chilling accumulation starts on 1 September of the autumn preceding budburst (Eq. 1). Forcing accumulation (Eq. 2) starts when the state of chilling (ChillState) equals the critical chilling requirement, "C_{crit}", (ChillState)≥ C_{crit}. Chilling temperatures affect the amount of forcing units required for budburst (F_{crit}) up to day "t_c" (Eq.3). Budburst occurs when the state of forcing (ForcState) ≥ F_{crit}	9 (a,b,c,d,e,co1, co2, C_{crit}, t_c)	(1)$ChState = \sum \frac{1}{1+e^{a(Tm-c)^2+b(Tm-c)}}$ (2)$ForcState = \sum \frac{1}{1+e^{d(Tm-e)}}$ (3)$Fcrit = co1 * e^{(co2 * Chstate)}$	x		
2	Sigmoidal heat accumulation starting on t_0 (calibrated calendar date)	4 (d, e, F_{crit}, t_0)	$ForcState = \sum \frac{1}{1+e^{d(Tm-e)}}$	x	x	x
3	Sigmoidal heat accumulation starting on previous phenophase	3 (d, e, F_{crit})	$ForcState = \sum \frac{1}{1+e^{d(Tm-e)}}$		x	x
4	Winkler degrees starting on t_0 (calibrated calendar date)	2 (F_{crit}, t_0)	$ForcState = \sum Tm - 10$		x	x
5	Winkler degrees starting on previous phenophase	1 (F_{crit})	$ForcState = \sum Tm - 10$		x	x
6	Degree days starting on previous phenophase (accumulation above a calibrated temperature threshold)	2 (F_{crit}, T_{thr})	$ForcState = \sum Tm - Tthr$		x	x
7	Degree days from t_0 (accumulation above a calibrated temperature threshold, starting on a calibrated calendar date)	3 F_{crit}, T_{thr}, t_0)	$ForcState = \sum Tm - Tthr$	x	x	x

F_{crit} Critical number of forcing units required to reach the phenophase of interest; *ForcState* number of accumulated forcing units; *ChState* number of accumulated chilling units; T_m mean daily temperature; T_{thr} threshold temperature above which forcing accumulation occurs; *a, b, c, d, e* curve shape parameters

selected (see Results). However, due to parameter overfitting, it was necessary to re-recalibrate the selected models with particular caution.

Calibration

To obtain robust and biologically sound models, we increased the amount of phenological information during model calibration, in the following three ways.

We used experimental information to restrict or fix potential parameter values to biologically realistic values Pouget (1968) studied the rates of phenological development at different temperatures in controlled environmental conditions. The "vitesse de debourrement" (speed of budburst, calculated as 1,000 / number of forcing days to 50% budburst) of dormant buds of different grapevine varieties was investigated at a range of forcing temperatures from 0 to 25°C. Although the varieties did not include Chardonnay, the responses had a similar general shape within the varieties, therefore we used the information pertaining to Merlot to estimate a rate of budburst that could be used in the models. The "vitesse de debourrement" was normalised to 25°C and fitted with a sigmoidal curve which was then employed as the forcing function in the phenological models (Fig. 2). In adopting this approach, we made three assumptions: first, that Chardonnay behaves similarly to the grapevine varieties studied by Pouget in 1968; second, that 25°C is close to the optimum temperature for grapevine phenological development; third, that the response to forcing temperature is the same from budburst to veraison. In addition, the range of values that could be adopted by the parameters C_{crit} (expressing the critical chilling requirement) and c (expressing the optimal chilling temperature) were restricted using information available in the literature. C_{crit} was restricted to between 5 and 100, as experimental evidence points to early dormancy release and low critical chilling requirements for grapevine (Pouget 1963; Dokoozlian 1999; Botelho et al. 2007). c was restricted to between 0 and 10, in accordance with the relevant experimental studies (Pouget 1963, 1968; Dokoozlian 1999; Botelho et al. 2007). These steps were aimed at obtaining a more biologically realistic representation of the chilling and forcing responses and at reducing the number of parameters to calibrate. In fact, the parameters d and e were fixed beforehand, preventing overfitting.

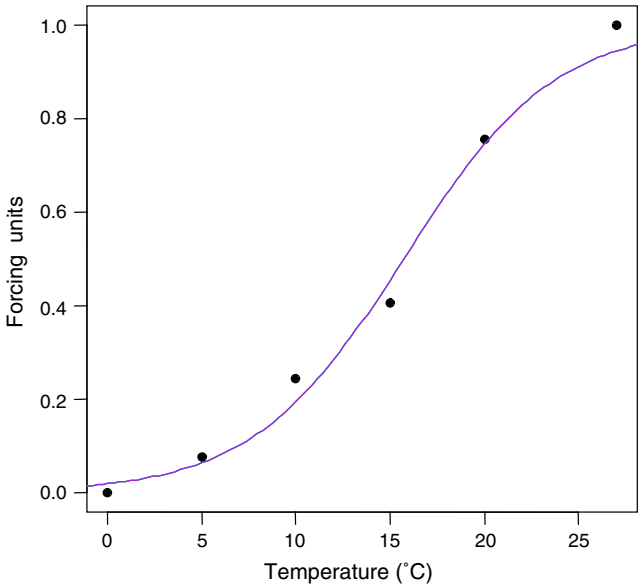


Fig. 2 Forcing function (Table 3 - budburst Eq. 2) fitted on data derived from the experimental study by Pouget (1968), as reported by Huglin and Schneider (1998). The study tested the "vitesse de debourrement" (1,000 / number of forcing days to 50% budburst) at different temperatures in different grapevine varieties. The data are normalised to 25°C

We simplified the budburst model The response to chilling temperature in the Unified model (Chuine 2000) is represented by a general curve defined by three parameters. However, since information on the shape of this response is not available and correct calibration of it is confounded by the absence of visible markers of chilling fulfilment, we simplified the general curve to a two-parameter bell curve peaking at 1 for temperature = c. This was obtained by setting b = 0 and multiplying the equation by 2 (Table 3, Eq. 1 of the budburst model).

We used a large phenological dataset for model calibration Based on the assumption that a long-term series contains more information on the response of phenology to a variety of weather conditions, we used the largest dataset available (Conegliano, 39 years) for model calibration.

Using these approaches, the number of parameters to be calibrated was reduced to six in the Unified model for budburst (a, c, C_{crit}, co1, co2, tc) and one for each of the flowering and veraison models (F_{crit}). The equations of the three models are shown in Table 3. While the starting date for the budburst model was 1 September, the flowering and veraison models were calibrated using simulated dates of previous phenophases (budburst for the flowering model and flowering for the veraison model) as starting dates. Thus, once calibrated, it was possible to join the three models together into a single, 8-parameter model which simulates the yearly dates of budburst, flowering and veraison with the input of mean daily temperatures. Each year, the timing of budburst is simulated with the budburst sub-model and used as the starting date for the simulation of flowering (flowering sub-model). Similarly, the simulated timing of flowering is used by the veraison sub-model as the starting date for the simulation of veraison. We refer to this model (and its sub-models) as "process-based" (PB) in contrast to other, more empirical models because it contains comparatively more assumptions on the physiology underlying phenological development.

Optimisation method

The optimisation method employed for the selection and calibration of the models was the Metropolis algorithm (Metropolis et al. 1953) which, in contrast to traditional numerical methods, such as the Downhill simplex and Newton's method, allows for a more effective exploration of the parameter space in functions with many local "minima", as is the case in phenological models (Chuine et al. 1998; Press et al. 1989).

Validation

The budburst, flowering and veraison models calibrated on the Conegliano dataset were validated on the remaining

Table 3 General forms of the phenological models applied to the Conegliano dataset for parameter estimation, after reduction of model parameters

Phenophase	Model name and description	No. of parameters	Equations
Budburst	Budburst model, derives from a simplified version of the Unified Model for Budburst (Chuine 2000).	6 (a, c, co1, co2, C_{crit}, t_c)	(1)$ChillState = \sum \frac{2}{1+e^{a(Tm-c)^2}}$ (2)$ForcState = \sum \frac{1}{1+e^{-0.26(Tm-16.06)}}$ (3)$Fcrit = co1 * e^{co2 * Chillstate}$
Flowering/veraison	Flowering/veraison models. Sigmoidal heat accumulation up to F_{crit}, starting on previous simulated phenophases.	1 (F_{crit})	$ForcState = \sum \frac{1}{1+e^{-0.26(Tm-16.06)}}$

C_{crit} Critical chilling requirement; F_{crit} critical number of forcing units required to reach the phenophase of interest; *ChillState* number of accumulated chilling units; *ForcState* number of accumulated forcing units; T_m mean daily temperature; *a, c, co1, co2* curve shape parameters

datasets (San Michele, Mezzocorona, Cembra and Carpeneto). Model performance was evaluated by "modelling efficiency" (Janssen and Heuberger 1995) and by the prediction improvement yielded by the models compared with the mean date of phenophases at Conegliano (external mean), which was considered as the null model. This improvement was expressed as the additional portion of variation explained by the model's prediction compared with the null model and is defined as:

$$AV = \frac{SS_{res1} - SS_{res2}}{SS_{res1}}$$

where SS_{res1} is the sum of squared residuals yielded by the null model and SS_{res2} is the sum of squared residuals yielded by the tested models.

Modelling efficiency (ME) is a measure analogous to the above-defined R_m^2, but refers to predictions made on a dataset external to the one used for calibration. Chuine et al. (1998) refers to it as "external validity". While ME expresses the portion of variation in the observation set accounted for by the model, AV is of interest for evaluating the ability of a model to yield better forecasts than the external mean. Like ME, when it assumes positive values (0 to 1), it indicates that the model is a better predictor than the mean, the contrary when it assumes negative values.

For further evaluation, we compared the accuracy of our process-based models with that of linear empirical models and Winkler degrees calibrated on the Conegliano dataset. For the linear models, regressions of mean monthly temperatures against the dates of budburst, flowering and veraison were performed on the Conegliano dataset. The best correlated mean temperatures were those of March for budburst, May for flowering and the period including May–July for veraison. Winkler degrees were calibrated for veraison, starting from the 1 April. These models were then applied to the temperatures of San Michele, Mezzocorona, Cembra and Carpeneto for external predictions and comparison.

As one of the aims of the models developed was to forecast plant phenology in a warming climate, we evaluated model accuracy in warm years. For each validation site, we selected the 3 years with the warmest May–July period, averaged the mean absolute errors ("warm-year index") given by the different models in those years, and compared them. The phenophase selected for this purpose was veraison, as its prediction by the PB model depends on the simulated dates of budburst and flowering and thus can be considered as a summary index. This index was useful for testing the accuracy of the model in extremely warm years, which are expected to occur more and more frequently, according to climate projections.

Finally, model bias was evaluated by plotting observed versus predicted dates and by calculating the mean bias error (MBE) (Mayer and Butler 1993).

$$MBE = \frac{\sum_{i=1}^{n} Xpred_i - Xobs_i}{n}$$

where *Xpred* stands for the predicted dates, *Xobs* for the observed dates and *n* for the number of cases.

A positive value of MBE indicates that, on average, the model overestimates actual observations, a negative value indicates that the model underestimates actual observations.

Bonferroni–Dunn tests were performed to compare groups of residuals (*Xpred–Xobs*) from each site in order to test for differences in model performance across sites. The Bonferroni–Dunn test allows multiple comparisons between groups and controls the family-wise error rate (Sheskin 2000).

Results

Model selection

The model that best explained the dates of budburst was based on the action of chilling temperatures for endodormancy release ("Unified model" for budburst in trees, Chuine 2000, model 1 among candidate models in Table 2). The models that best described flowering and veraison considered phenophase attainment as the result of the accumulation of heat and calculated forcing units through a sigmoidal function (model 3 in Table 2). However, the lack of consistency among parameter values fitted to different datasets suggested that overfitting might have occurred. In fact, due to the small variation in the observations and temperature series, the parameters might have adapted to noise in the data and taken unrealistic values.

Model calibration

The "informed" calibration of the models selected during the first phase of the study resulted in three PB (sub-) models, one for each of the phenophases of interest. The functions calibrated by the budburst model are plotted in Fig. 3. The parameter estimates for each of the models are given in Table 4.

The portion of variation (R_m^2) in the Conegliano phenological series explained by the budburst, flowering and veraison models were 0.36, 0.79 and 0.69, respectively. The *F* test performed on the results indicated that in all cases the models were significant ($P < 0.05$).

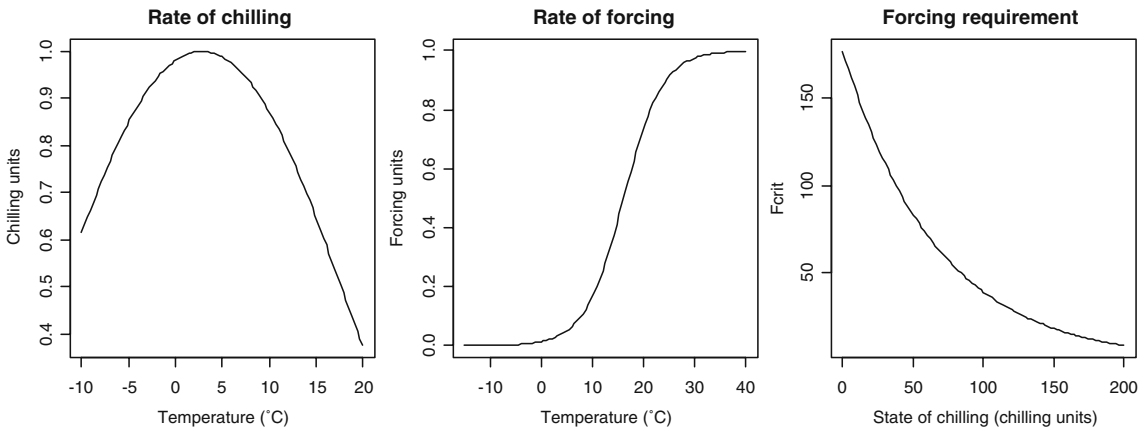


Fig. 3 Modified version of the "Unified model" by Chuine (2000), as calibrated on the Conegliano dataset: response to temperature (°C) during the various stages of dormancy. For the general equations of the model, see Table 3, for parameter values, see Table 4

Model validation

When the PB models were applied to external datasets for validation, their prediction errors were always smaller than those made with the external mean used as a benchmark predictor (mean date of the phenophases of interest in Conegliano) (Table 5). This implies that the models were able to account for part of the variation in the external observations, unexplained by the mean date of phenophase occurrence at the site of calibration. Some differences in model performance and among stations were observed. For example, in contrast to AV, ME was very low for veraison in Carpeneto. Among the PB models applied to the three phenophases, the veraison model was the better performing, with an average AV of 52%, compared to averages of 41% and 33% for the flowering and budburst models, respectively. Comparison between the different models shows that process-based models on average yielded lower MAE, higher ME and explained a higher portion of additional variation (AV) compared to linear regression models and Winkler degrees (Table 5). This was especially so for flowering and veraison, while the improvement was less evident for budburst.

The "warm-year index", calculated over the three warmest years for each site, indicated that PB models yielded markedly better predictions in warm years compared to linear models. This occurred at all sites, as shown in Table 6.

The plots of observed versus predicted dates revealed that the dates of flowering and veraison were overestimated by the PB models at Cembra, the only "mountain" site (Fig. 4). For this site, the PB model predicted flowering and veraison later than actual observations 85 and 91% of the time, respectively. On the other hand, phenological events at Carpeneto, the warmest of the four sites, tended to be underpredicted by the PB models, while San Michele and Mezzocorona, which have similar temperatures, displayed an intermediate pattern. The MBEs at each site ordered by mean temperature are shown in Fig. 5. The large positive MBEs at Cembra contrast with those of the other stations sited at lower altitudes and with a higher mean temperature. The Bonferroni-Dunn test confirmed that the flowering and veraison sub-models in Cembra yielded significantly different residuals ($P<0.05$) compared to the other sites, except for flowering in Mezzocorona ($P=0.57$).

Table 4 Parameter values for the best-fit models for each phenophase

Model	Parameter estimates
Budburst	a: 0.005
	c: 2.800
	co1: 176.260
	co2: −0.015
	C_{crit}: 78.692
	t_c: 234
Flowering	F_{crit}: 24.710
Veraison	F_{crit}: 51.146

C_{crit} Critical chilling requirement; F_{crit} critical number of forcing units required to reach the phenophase of interest; *a, c, co1, co2* curve shape parameters; t_c last day of effect of chilling on F_{crit} (starting from 1 September)

Discussion

Calibration

Process-based models are usually developed using available information on the processes involved in the system under consideration. Previously established relationships, hypothesised new relationships or, more commonly, a combination of both approaches may be integrated into them. When only the general behaviour of the processes being modelled is

Table 5 Goodness of fit of the phenological models calibrated on the Conegliano dataset, and their predicting performance

Phenophase	Model	Calibration		Prediction										Average AV	Average ME	Average MAE
		Conegliano		San Michele			Mezzocorona			Cembra			Carpeneto			
		R_m^2	MAE	AV	ME	MAE	AV	ME	MAE	AV	ME	MAE	AV	ME	MAE			
Budburst	L	0.33	4.9	0.47	0.49	4.6	0.15	0.15	5.8	NA			0.28	0.28	5.6	0.30	0.31	5.33
	PB	0.36	4.8	0.48	0.49	4.5	0.27	0.27	5.1				0.23	0.22	5.7	0.33	0.33	5.10
Flowering	L	0.53	3.5	0.38	0.07	4	−0.01	−0.28	4.8	0.34	0.08	5.7	0.17	0.11	4.2	0.22	−0.01	4.68
	PB	0.79	3.1	0.41	0.13	3.29	0.34	0.16	4	0.50	0.29	5.3	0.39	0.34	3.6	0.41	0.23	4.05
Veraison	W	0.55	4.7	0.28	0.01	5.5	0.05	−0.58	4.7	0.31	−0.08	6.9	0.29	−0.42	5.6	0.23	−0.27	5.68
	L	0.59	4.5	0.15	−0.17	5.12	0.24	−0.26	4.8	0.56	0.32	5.6	0.27	−0.47	5.1	0.31	−0.15	5.16
	PB	0.69	3.8	0.46	0.26	4.41	0.51	0.19	3.7	0.64	0.44	5.2	0.48	−0.04	4.3	0.52	0.21	4.40

L Linear regression model; *PB* process-based model; *W* Winkler degrees; R_m^2 portion of explained internal variance; *MAE* mean absolute error (°C); *AV* portion of additional variance explained by the model compared to the external mean; *ME* modelling efficiency

known or assumed, its parameters can be obtained by calibrating the model on a suitable dataset. Parameters are estimated through optimisation of the model fit, so that the model can be considered as being only partially process-based (Cox et al. 2006). This was our case, as it was necessary to estimate parameter values using numerical techniques, but also because we did not make any hypotheses on the actual molecular processes underlying development. Nonetheless, the model proposed here makes an attempt at describing causal relationships between triggers and phenology at a macroscopic, whole-plant level using experimental findings, and is thus comparatively more process-oriented than other widely-used models, such as Winkler degrees.

The models selected in a preliminary phase were simplified, and, to avoid model overfitting, some of the parameters were fixed (or their ranges were limited) according to experimental evidence. Despite the restraints imposed by these measures, the three models described their calibration dataset well. The high internal validities (R_m^2) of the flowering and veraison models were notable, given that they were based on experimental findings on grapevine varieties other than Chardonnay.

The general framework employed for budburst simulation was developed by Chuine (2000) and named the "Unified model". It was developed for the budburst of trees and consists in a set of general equations accounting for the most widely-used models of budburst of perennial species (Chuine 2000). Among these, the model that best describes the data set under study is inferred after calibration, according to best fit parameter estimates. These indicate the model structure, defined by the period of action of chilling and forcing temperature and the shape of the functions linking temperature with phenological development. The chilling function plotted in Fig. 2 shows optimal efficiency around 2.8°C, but the range of active chilling temperature is quite wide. On the other hand, the function relating the critical forcing requirement with the amount of accumulated chilling units shows a decrease as chilling progresses after the attainment of C_{crit} (occurring, on average, around the end of December, according to the simulation). The fitted model can thus be ascribed to a parallel model (see Landsberg 1974; Hänninen 1987, 1990), where forcing temperatures are active simultaneously with chilling temperatures.

According to the parameters fitted to the budburst model, the critical state of chilling is reached at 79 chilling units, meaning that 79 days at optimal chilling temperature are required to fulfil chilling. This critical chilling requirement may seem high, given the literature on the subject. Botelho et al. (2007) report a chilling requirement of about 336 h at temperatures below 6°C for Cabernet Sauvignon, which was fulfilled in about 2 weeks. However, in our case,

Table 6 Absolute errors yielded by the three phenological models for the prediction of veraison in the three warmest years for each station, and their average ("warm year index")

Model	San Michele			Mezzocorona			Cembra			Carpeneto		
	W	LM	PB	W	LM	PB	W	LM	PB	W	LM	PB
Warmest year	11	12	11	4	1	3	7	3	4	0	6	0
2nd warmest year	13	14	12	3	24	1	4	6	4	10	12	9
3rd warmest year	1	4	1	5	6	2	5	6	4	10	11	9
Average (warm year index)	8.1	10.1	7.8	3.6	10.3	1.8	5	5.2	4.1	6.8	9.7	6

L Linear regression model; *PB* process-based model; *W* Winkler degrees

the calibrated chilling curve considers temperatures above 10°C also to be effective. In September, when the model starts to compute chilling units, mean daily temperatures (usually between 10 and 20°C) are thus considered effective for "chilling accumulation". While this does not make much sense in physiological terms, as endodormancy is typically broken by temperatures below 10–12°C (Pouget 1968; Dokoozlian 1999), it may indicate that mild temperatures in the autumn contribute to early dormancy (and subsequent release), while warm temperatures above 20°C delay it. The budburst delaying effect of high temperatures during dormancy induction was also reported by Heide (2003) for *B. pubescens*. According to our model, after fulfilment of the critical chilling requirement (C_{crit}), which is attained on average at the beginning of December, low temperatures act on the system mainly by decreasing the forcing requirement for budburst, while later, warmer conditions affect the system by increasing the sum of accumulated forcing units. "Chilling" temperatures are effective until the day "tc", calibrated as the calendar date 21 April, which is usually after budburst. This implies that temperatures in the range of 10–20°C have a dual effect (decreasing the critical forcing demand, increasing the sum of forcing units) until the day of budburst.

The form taken by the budburst model functions unfortunately does not provide exact information on the real chilling requirements for Chardonnay, but it highlights the dynamicity of the modelled system and possibly points to a constant component in the rate of phenological development, as all temperatures contribute to development (either by contributing to fulfilment of C_{crit}, or by decreasing F_{crit}, or by increasing the state of forcing) during all stages of dormancy. In order to calibrate correctly these relationships, a dataset containing observations taken in extremely warm years would be useful. Experiments in controlled conditions on the effect of chilling temperature on dormancy are crucial for correct parameterisation of chilling rates and for a correct representation of the processes leading to budburst. Budburst models to be used in climate change projections should consider the effect of chilling temperatures, as in a much warmer climate the chilling requirement of grapevine might not be fulfilled, resulting in incomplete budburst (García de Cortázar-Atauri et al. 2009).

Validation

While the exact response to chilling was probably not caught by the model fit, due to non-limiting conditions for chilling in the calibration dataset, the budburst model seems to have caught the behaviour of the system better than other models. This is evidenced by its ability to explain more variance than the external mean date of budburst (AV, 36%) and to account for some of the variance in the observed data (ME, 31%). Given the current difficulties in developing budburst models for grapevine which are both physiologically meaningful and have external validity, this is a significant result. For example, García de Cortázar-Atauri et al. (2009) tried a set of different models to calculate budburst in different varieties of grapevine and found that the best predictive model (BRIN model) considered the action of chilling temperature for dormancy release and used a base temperature of 5°C. However, following statistical testing, he concluded that the reason for the better performance of the model was the use of the 5°C temperature threshold, rather than the simulation of dormancy release by chilling temperatures. Similarly, Bindi et al. (1997) found no model more accurate than the average number of days between 1 January and budburst. In any case, using the simulated date of budburst as a starting day for the flowering model did not hamper the prediction of flowering dates at any of the four stations. Indeed, prediction of the timing of flowering yielded low MAEs on average. This was an important result, as the date of flowering is known to be an important predictor of the subsequent phenophases (Blouin and Guimberteau 2000).

On average, the veraison model yielded good predictions of this phenophase. This model performed equally well at all stations, as opposed to the linear regression model, which performed poorly at San Michele and in general yielded higher MAEs and lower AVs. It also performed better than the model based on Winkler-degrees, which was

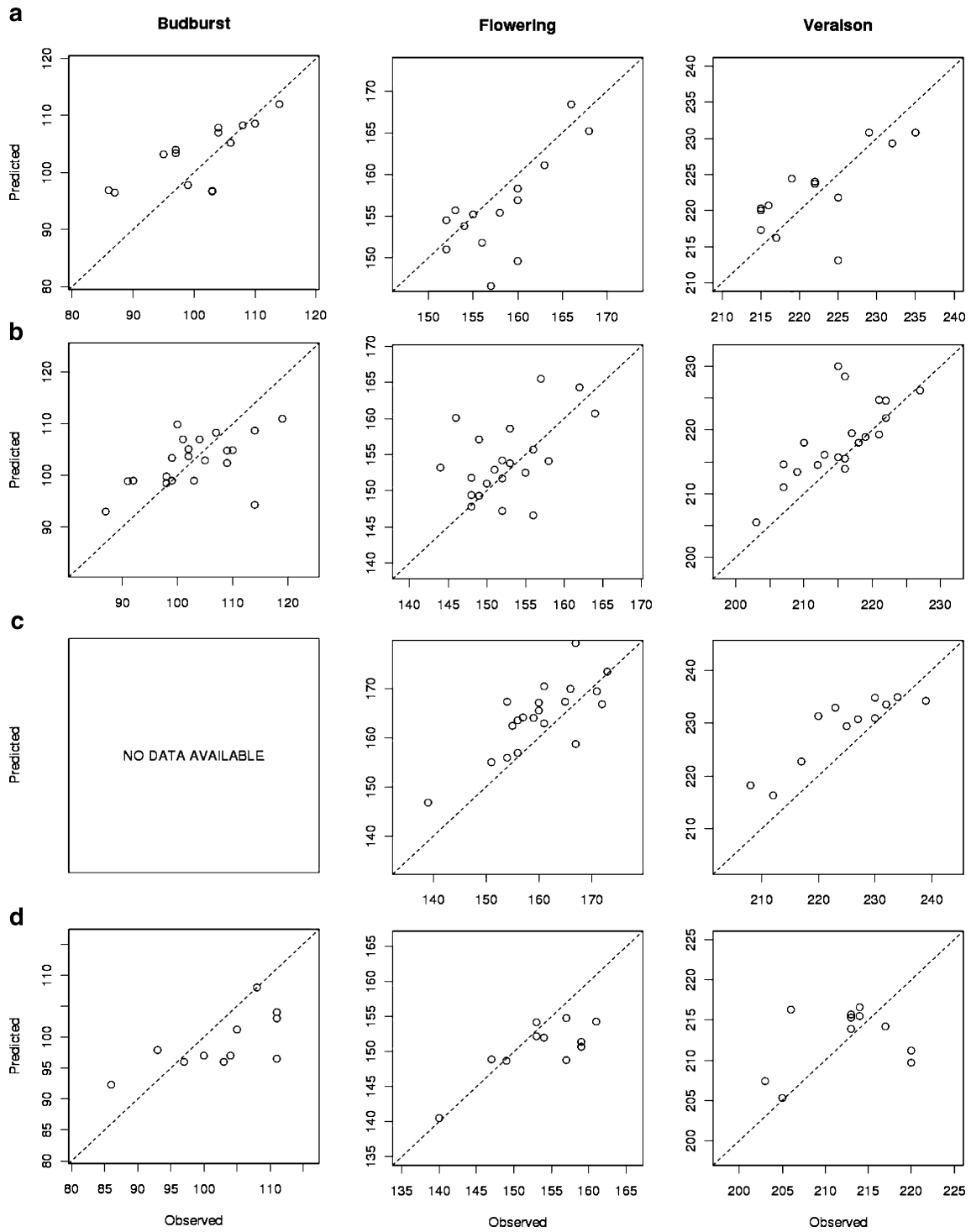


Fig. 4 Plots of observed versus predicted dates for each site and phenophase available, by the PB models. **a** San Michele, **b** Mezzocorona, **c** Cembra, **d** Carpeneto

unable to predict veraison at Cembra (negative AV) and performed poorly at Mezzocorona (0.04 AV and 4.7 MAE). The better performance of the PB veraison model is probably due to the fact that, unlike linear regression and Winkler degrees, it starts its computation from the simulated dates of flowering. In fact, the timing of flowering is more sensitive to changes in meteorological conditions than veraison, as is also shown by the greater variance in the budburst–flowering interval compared to the flowering–veraison interval. Genetic studies have shown that the time lag between flowering and veraison is a heritable trait and thus responds to an inner clock, only partially affected by

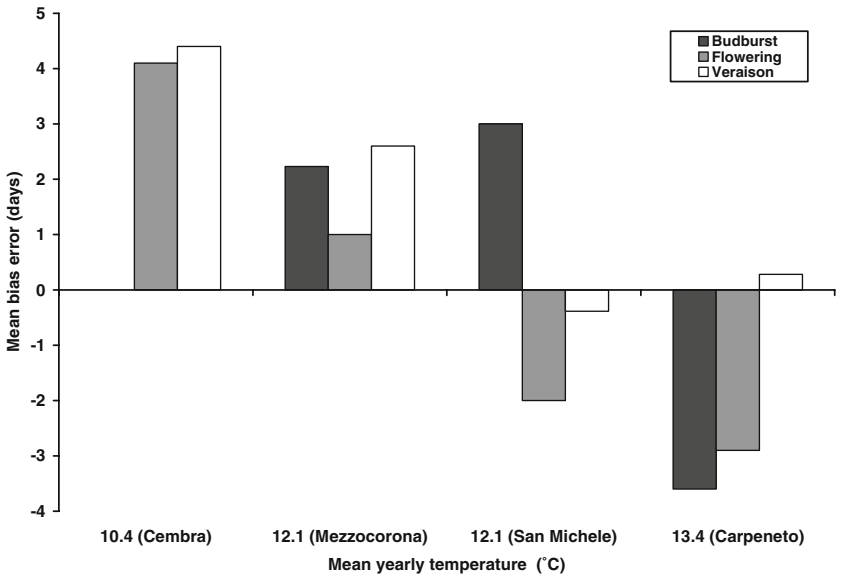


Fig. 5 Mean bias error for the prediction of budburst, flowering and veraison at each site, ordered by mean yearly temperature. The relevant sites are given in parentheses

temperature (Costantini et al. 2008). This implies that a good model of flowering is fundamental for accurate prediction of the subsequent phenophases. Another advantage of the PB veraison model was its use of a sigmoidal instead of a linear forcing function. This function type is more suitable for describing the response to warm, supra-optimal temperatures for growth. It has been shown that the optimal growing temperatures for grapevine are in the high 20s, above which growth rate levels out or even decreases (Ferrini et al. 1995). This pattern is not considered by degree day models but might become significant in particularly warm years, such as those predicted by climate change scenarios. The consistently more accurate predictions of veraison by the PB model in the three warmest years for each station ("warm-year index") show the advantage of a correct representation of the response to warm temperatures. Interestingly, while the overall MAE was lower for the linear regression model than for the model based on Winkler degrees, the "warm-year index" showed a better performance by the Winkler-degree model. In fact, Winkler degrees might offer better predictions at warm sites compared to linear models. However, both models were outperformed by the PB model for the stations under study.

Visual evaluation of the scatterplots for observations versus predicted dates showed that, despite the good external accuracy of the PB model, the dates of flowering and veraison tended to be overestimated at the Cembra site. The MBE for Cembra confirms this qualitative evaluation. The bias was consistent, occurring in most of the years in the data series, and might be due to a systematic error in the temperature records or to differences in the observational protocol. It might also be due to environmental differences unaccounted for by the model. Indeed, Cembra is the only "mountain" site among those studied and its mean May–July temperature is 1.3–2.5°C lower than at the other sites. The phenological rate of development of apple in the same climatic conditions as the present study has been reported to be comparatively faster at higher altitude, even within the same apple clone (Rea and Eccel 2006). A common garden experiment on tree species from different altitudes revealed an altitudinal cline in the dates of budburst and senescence, and concluded that altitude leads to genetic differentiations among populations within the same species, resulting in different sensitivities to temperature (Vitasse et al. 2009). In our case, we are dealing with genetically homogeneous clones whose phenological response should have the same genetic basis. Nonetheless, the response to temperature appears to be significantly different at Cembra compared to the other sites. This difference might be due to phenotypic adaptation to a lower mean temperature. Kramer (1995) showed that along latitudinal and temporal gradients the phenotypic responses to temperature in the time of leaf unfolding varied by a similar magnitude in beech clones and in genetically different trees (2.0 and 2.4 days per °C for genetically different trees and clones, respectively). The nature of this plasticity should be further investigated in order to provide more insight into the mechanisms regulating phenology at mountain sites or in cool regions. Understanding them would be fundamental for more efficient crop modelling and management, for the selection of cultivars to be grown at elevated sites, and for predicting the impact of climate change on viticulture in general. In fact, this "phenological resilience" could be a key feature for short-term crop adaptation.

To obtain a truly general model for Chardonnay, and to prevent biased estimations of phenophases at mountain sites, it would be very useful to establish a relationship between the mean temperature at the growing site and the

parameters of the present model. The already high accuracy of the veraison model could be further improved. Without the systematic error found in the prediction of veraison in Cembra, the AV rises to 85% and the MAE is only 3.3 days. Such accuracy would be very useful even for the immediate planning of agricultural practices, such as pest control. The short-term prediction of phenophases would, of course, depend on a reliable weather forecast providing weekly average temperature forecasts for up to 2–3 weeks ahead. In order to achieve this robustness, more phenological data from different altitudes and climate conditions are necessary. It is crucial that these data be collected with rigour and be supported by precise and accurate temperature records.

Conclusions

The aim of this study was to obtain a spatially robust model for predicting the phenophases of budburst, flowering and veraison in *V. vinifera* cv. Chardonnay. A preliminary study and model selection suggested the type of sub-models to be used for each of these phenophases and indicated a set of measures that would prevent overfitting during model calibration, besides obtaining a more physiologically meaningful process-based model. The resulting model showed good internal validity and good accuracy when predicting phenophases in external datasets from Trentino and other northern Italian regions. Model performance was especially high for the phenophase of veraison, and comparison with other models showed that it was a better predictor of phenology, even in extremely warm years.

The modelling study highlighted a different phenological behaviour for the only mountain station, Cembra. We hypothesised that a lower mean temperature could lead to phenotypical adaptation of growth rates, which is not generally accounted for by phenological models. Quantification of this relationship could further improve the performance of the model. In order to do so, it would be useful to test the phenological responses of clones from different climate regimes during experimental trials and compare phenological data from different altitudes.

Nonetheless, the model in its present form outperforms both linear regression and Winkler degree models, and yields reasonably accurate predictions of the phenophases considered. It could be used to plan agricultural practices in the short term and to predict the impact of climate change on viticulture in the long term.

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